Mixed Factoring: Choosing and Combining Methods Answer Key

Algebra 1 Polynomials

Quick Answers

1. $3x\,(2x-5)$
2. $(x+4)\,(x+5)$
3. $(x-8)\,(x+8)$
4. $(x+3)\,(3x-9)$
5. $(x+3)\,(2x+1)$
6. $(x-7)\,(x+4)$
7. $(2x-3)^2$
8. $2x\,(x-3)\,(x+3)$
9. $(2x-3)\,(3x+1)$
10. $2x^2+8x+8,\ 2\,(x+2)^2$
11. $(x-3)\,(x+3)\left(x^2+9\right)$
12. $4x\,(x-2)\,(x+3)$

Curriculum

Level: Algebra 1 Polynomials
HSA-APR.A.1 – *problems 8, 10*
HSA-SSE.B.3 – *problems 1, 2, 3, 4, 5, 6, 7, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 14 tagged checks.

1. Factor completely: $6x^2-15x$.

Step 1. Checklist step one: is there a common factor? The coefficients 6 and 15 share 3, and both terms carry an x, so the greatest common factor is $3x$.
Step 2. $6x^2-15x=3x(2x-5)$.
Step 3. The bracket has two terms but no subtraction of two squares, and it is degree one, so nothing factors further. Technique: GCF only. $\boxed{3x(2x-5)}$

2. Factor completely: $x^2+9x+20$.

Step 1. No common factor. Three terms with leading coefficient 1, so look for a factor pair of 20 that adds to 9.
Step 2. 4 and 5: $4\cdot 5=20$ and $4+5=9$ ✓
Step 3. $x^2+9x+20=(x+4)(x+5)$. Technique: monic trinomial. $\boxed{(x+4)(x+5)}$

3. Factor completely: x^2-64.

Step 1. No common factor. Two terms with a subtraction sign between them: test the difference of squares.
Step 2. $x^2=(x)^2$ and $64=(8)^2$, so $A=x$ and $B=8$.
Step 3. $A^2-B^2=(A-B)(A+B)$ gives $(x-8)(x+8)$. Expanding back, the middle terms $+8x$ and $-8x$ cancel ✓ Technique: difference of squares. $\boxed{(x-8)(x+8)}$

4. Factor completely: $3x^2 - 27$.
Step 1. It looks like a difference of squares, but $3x^2$ is not a perfect square, so the pattern does not apply yet. Checklist step one first: both terms are divisible by 3.
Step 2. $3x^2 - 27 = 3(x^2 - 9)$.
Step 3. Now the bracket *is* a difference of squares with $A = x$ and $B = 3$, so $x^2 - 9 = (x-3)(x+3)$ and the complete factorization is $3(x-3)(x+3)$. Technique: GCF, then difference of squares. $\boxed{3(x-3)(x+3)}$

5. Factor completely: $2x^2 + 7x + 3$.
Step 1. No common factor (3 is odd), and the leading coefficient is 2, not 1 — so factor by grouping on the product of the outer coefficients.
Step 2. That product is $2 \cdot 3 = 6$, and we need a factor pair of 6 adding to 7: those are 6 and 1. Split the middle term: $2x^2 + 6x + x + 3$.
Step 3. Group: $2x(x+3) + 1(x+3) = (2x+1)(x+3)$. Technique: non-monic trinomial by grouping. $\boxed{(2x+1)(x+3)}$

6. Flower bed of area $x^2 - 3x - 28$ square feet.
Step 1. A rectangle's area is length times width, so factoring the area into two brackets is exactly what produces the two side lengths. No common factor, leading coefficient 1: look for a factor pair of -28 that adds to -3.
Step 2. The constant is negative, so the pair has opposite signs and the larger one is negative: -7 and $+4$ give $-7 \cdot 4 = -28$ and $-7 + 4 = -3$ ✓
Step 3. $x^2 - 3x - 28 = (x-7)(x+4)$, so the bed measures $x-7$ feet by $x+4$ feet. Technique: monic trinomial. $\boxed{(x-7)(x+4)}$

7. Factor completely: $4x^2 - 12x + 9$.
Step 1. No common factor (9 is odd). Three terms whose first and last are perfect squares — test the perfect-square pattern before trying pairs.
Step 2. $4x^2 = (2x)^2$ and $9 = (3)^2$, so $A = 2x$ and $B = 3$. The pattern demands a middle term of $2AB = 2 \cdot 2x \cdot 3 = 12x$, and the trinomial has $-12x$: the size matches and the sign is negative.
Step 3. A negative middle term means the bracket is a difference: $4x^2 - 12x + 9 = (2x-3)^2$. Technique: perfect-square trinomial. $\boxed{(2x-3)^2}$

8. Factor completely: $2x^3 - 18x$.
Step 1. Two terms, but neither is a perfect square as written — and both are divisible by 2 and both contain x. The greatest common factor is $2x$.
Step 2. $2x^3 - 18x = 2x(x^2 - 9)$.
Step 3. The bracket is a difference of squares with $A = x$, $B = 3$: $x^2 - 9 = (x-3)(x+3)$. Complete: $2x(x-3)(x+3)$. Technique: GCF, then difference of squares.
$\boxed{2x(x-3)(x+3)}$

9. **Factor completely:** $6x^2 - 7x - 3$.

Step 1. No common factor, leading coefficient 6: grouping again. The product of the outer coefficients is $6 \cdot (-3) = -18$.

Step 2. A factor pair of -18 adding to -7 is -9 and $+2$. Split the middle term: $6x^2 - 9x + 2x - 3$.

Step 3. Group: $3x(2x-3)+1(2x-3) = (2x-3)(3x+1)$. Expanding back gives $6x^2+2x-9x-3 = 6x^2-7x-3$ ✓ Technique: non-monic trinomial by grouping. $\boxed{(2x-3)(3x+1)}$

10. **Error analysis: is** $2x^2 + 8x + 8 = (2x+4)(x+2)$ **complete?**

Step 1 (part a). Multiply the student's answer out: $(2x+4)(x+2) = 2x^2+4x+4x+8 = 2x^2+8x+8$. It really does equal the original, so expanding alone cannot expose the fault — this is why "completely" matters.

Step 2 (part b). The first bracket still has a common factor: $2x+4 = 2(x+2)$. The student skipped checklist step one, so a factor of 2 was left buried inside a bracket instead of pulled to the front.

Step 3 (part c). Take the GCF out first: $2x^2+8x+8 = 2(x^2+4x+4)$, and x^2+4x+4 is a perfect square, $(x+2)^2$. So the complete factorization is $2(x+2)^2$.

Manual review (part b): accept any sentence saying that $2x+4$ still contains a common factor of 2, or equivalently that the answer is a correct product but not fully factored.

$\boxed{2x^2+8x+8 \quad \text{and} \quad 2(x+2)^2}$

11. **Factor completely:** $x^4 - 81$.

Step 1. No common factor. Two terms with a subtraction sign, and both are squares: $x^4 = (x^2)^2$ and $81 = (9)^2$.

Step 2. Difference of squares with $A = x^2$, $B = 9$: $x^4 - 81 = (x^2-9)(x^2+9)$.

Step 3. Check each bracket again. $x^2 - 9$ is itself a difference of squares, $(x-3)(x+3)$; x^2+9 is a *sum* of squares and does not factor over the integers, so the work stops there. Technique: difference of squares, twice. $\boxed{(x-3)(x+3)(x^2+9)}$

12. **Synthesis challenge: factor completely** $4x^3 + 4x^2 - 24x$.

Step 1. Checklist step one: the coefficients 4, 4 and 24 all share 4, and every term carries an x, so the greatest common factor is $4x$.

Step 2. $4x^3+4x^2-24x = 4x(x^2+x-6)$. The bracket now has leading coefficient 1, which is the easy trinomial case — taking the GCF out first is what turned a messy cubic into it.

Step 3. A factor pair of -6 adding to $+1$ is $+3$ and -2, so $x^2+x-6 = (x+3)(x-2)$ and the complete factorization is $4x(x+3)(x-2)$. Technique: GCF, then monic trinomial.

$\boxed{4x(x+3)(x-2)}$